Enzymes

Exam code: 0610 & 0970

Download PDF

Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
Updated on 15 May 2025

Did this video help you? Yes No

Help me understand

What are enzymes?

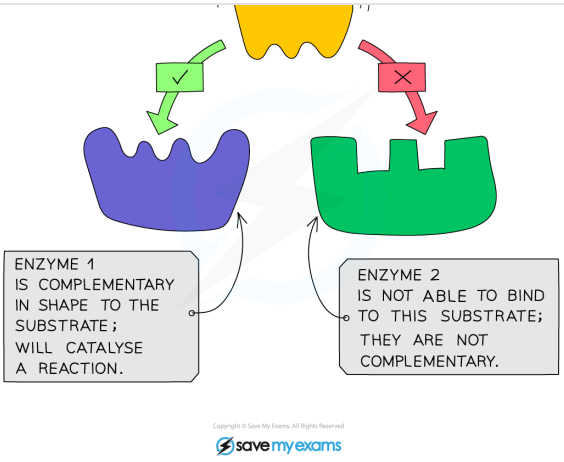
- Enzymes are:
  - **Catalysts** that **speed up** the rate of a chemical reaction **without being changed** or used up in the reaction
  - **Proteins**
  - **Biological catalysts** (biological because they are **made in living cells**, catalysts because they speed up the rate of chemical reactions without being changed)
  - Necessary to all living organisms as they **maintain reaction speeds** of all metabolic reactions (all the reactions that keep an organism alive) at a rate that can **sustain life**
    - For example, if we did not produce digestive enzymes, it would take around 2 - 3 weeks to digest one meal; with enzymes, it takes around 4 hours

Stuck on something?

Ask questions and get help with exactly what you're revising.

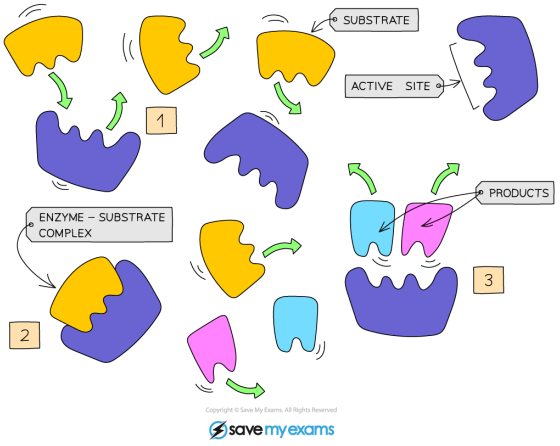
Got it

How do enzymes work?



Enzyme substrate specificity

- Enzymes are **specific** to one particular substrate (molecule/s that get broken down or joined together in the reaction) as the enzyme is a complementary shape to the substrate
- The product is made from the substrate(s) and is released



Enzyme specificity: lock and key model of enzyme activity

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder →

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.